

**Bupropion,  $\alpha 7$  nicotinic signaling, dissociation and seizure susceptibility:  
two linked hypotheses with preliminary docking evidence**

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## **Abstract**

Bupropion inhibits  $\alpha 7$  nicotinic acetylcholine receptors in experimental preparations, while its metabolites have incompletely characterized  $\alpha 7$  pharmacology. We develop two related hypotheses from a focused critical review and an existing stereoisomer-resolved docking study. Hypothesis 1 proposes that, in susceptible individuals with sufficient target exposure,  $\alpha 7$  inhibition contributes to bupropion-associated depersonalization or derealization by disturbing interneuron-dependent sensory selection and the integration of bodily and external signals. Hypothesis 2 proposes that a shared loss of cholinergic support for inhibition can produce sensory-processing disturbances at lesser perturbation and increase seizure susceptibility at greater perturbation, provided both outcomes involve the same susceptible circuit. Evidence supports individual links, including parent-drug  $\alpha 7$  inhibition, circuit-specific cholinergic recruitment of inhibition, and dissociative experiences during some focal seizures. The complete drug–receptor–circuit–symptom chain remains untested. Null findings, opposite effects of  $\alpha 7$  activation, non- $\alpha 7$  pharmacology and depersonalization imaging findings constrain the proposals. Docking identifies candidate metabolite locations but supplies neither potency nor human target engagement. We specify discriminating predictions, conditions under which the hypotheses separate, and observations that would reject each mechanism.

**Keywords:** bupropion; hydroxybupropion;  $\alpha 7$  nicotinic acetylcholine receptor; depersonalization; derealization; seizure susceptibility.

## **The two hypotheses**

**Hypothesis 1 — an  $\alpha 7$  contribution to bupropion-associated dissociation.** In a susceptible subgroup, bupropion—and only those metabolites subsequently shown to inhibit  $\alpha 7$  at relevant unbound concentrations—reduces cholinergic recruitment of inhibitory interneurons in sensory and associative circuits. The resulting disturbance of sensory selection contributes to altered integration of external information with bodily and emotional signals, producing depersonalization and/or derealization.

**Hypothesis 2 — a conditional shared mechanism for sensory disturbance and seizure susceptibility.** In circuits where  $\alpha 7$  activity normally strengthens effective inhibition, increasing inhibition of that receptor by bupropion or an active metabolite progressively weakens network control. Lesser disturbance can alter sensory processing; greater disturbance can facilitate pathological synchronization and lower seizure threshold. This shared-mechanism hypothesis predicts linked,  $\alpha 7$ -dependent intermediate changes across the two outcomes. It does not assume that every dissociative experience is epileptic, that symptoms must progress to seizures, or that all bupropion seizures use this mechanism.

These are distinct claims. Hypothesis 1 could hold while Hypothesis 2 fails, for example if dissociative symptoms involve  $\alpha 7$  but convulsions arise mainly from another target. A seizure-specific  $\alpha 7$  contribution would likewise not establish the dissociation hypothesis.

### **1. Clinical starting point and definitions**

The current Wellbutrin XL label lists depersonalization and derealization among postmarketing nervous-system events, while explicitly stating that voluntary reports cannot reliably establish frequency or causality. Its seizure warning provides a separate clinical motivation. A list of reported events does not demonstrate that the events share a receptor mechanism. (1)

Here, **depersonalization** denotes subjective detachment from oneself or one's body; **derealization** denotes altered reality or familiarity of the surroundings. Medication-associated symptoms, persistent primary depersonalization–derealization disorder, and brief seizure-associated experiences must be analyzed separately. **Sensory gating** is also not a single endpoint: auditory P50 suppression, visual P1 attention effects and acoustic-startle prepulse inhibition (PPI) measure different processes. None is a substitute for a validated dissociation assessment.

## **2. What is established at the receptor level**

Parent bupropion has experimental  $\alpha 7$  inhibitory activity. Slemmer et al. compared several nicotinic subtypes; Alkondon and Albuquerque found no inhibition of the  $\alpha 7$ -like response at 1  $\mu\text{M}$  in their native-slice experiment, despite effects on other response classes. Papke et al. measured human  $\alpha 7$   $\text{IC}_{50}$  values of  $46 \pm 7$   $\mu\text{M}$  during coapplication and  $16 \pm 3$   $\mu\text{M}$  after 30 minutes of bath exposure. The protocols also differed in acetylcholine concentration, preventing a purely time-based explanation of the difference. Vázquez-Gómez et al. reported a 54  $\mu\text{M}$  calcium-response  $\text{IC}_{50}$  and native-neuron inhibition; Duarte et al. reported inhibition of rat  $\alpha 7$ -containing currents at 20 and 50  $\mu\text{M}$ . These results demonstrate assay-dependent inhibition, not therapeutic human receptor occupancy. (2, 3, 4, 5, 6)

The metabolite evidence requires a different statement. In the directly inspected Lukas et al. functional panel, S,S-hydroxybupropion was more potent than parent at  $\alpha 4\beta 2$ , while R,R-hydroxybupropion was less potent. The panel did not contain  $\alpha 7$ . (7)

Comparative assay results are summarized in Table 1.

The six core hydroxy-, threo- and erythrohydrobupropion stereoisomers should therefore remain separate experimental candidates. Minor metabolites and conjugates require their own

functional and brain-exposure evidence; structural identification in urine does not establish CNS action. Synthetic analogues, deuterated compounds and unresolved glucuronide candidates are not interchangeable with established circulating metabolites.

### **3. Hypothesis 1: from $\alpha 7$ inhibition to dissociative symptoms**

#### ***Rationale***

Nicotinic receptors can recruit GABAergic inhibition. Human cortical and rat hippocampal recordings show nicotinic control of interneuron output, with either inhibition or disinhibition depending on the connection being activated. Rat basolateral-amygdala recordings show a condition in which  $\alpha 7$  activity favors inhibition over excitation. Townsend et al. found increased inhibitory currents and reduced pyramidal firing during  $\alpha 7$  agonist application; MLA increased evoked firing. These provide mechanistic plausibility for loss of cholinergic inhibition, but not proof of bupropion acting through that route in a patient. (10, 11, 12, 13)

Clinical sensory evidence is mixed. The small DMXB-A trial and a tropisetron trial reported P50 improvements in schizophrenia; an  $\alpha 7$  positive-allosteric-modulator trial did not. Tropisetron also acts at 5-HT<sub>3</sub> receptors, and results in schizophrenia do not establish effects in depersonalization. A CHRNA7 promoter study found no association of the tested variant with P50 suppression or PPI, despite association with schizophrenia and a startle finding. (14, 15, 16, 17)

The most relevant symptom studies support altered processing without identifying  $\alpha 7$ . Schabinger et al. found reduced early suppression of unattended visual stimuli in depersonalization. Dale et al.'s small depersonalization subgroup broadly resembled controls on PPI, whereas abnormalities concerned a separate dissociative-identity group. Bupropion did not

disrupt PPI in Mori et al.'s mouse experiment. These findings require endpoint-specific predictions rather than a blanket claim of failed filtering. (18, 19, 20)

### ***The missing bridge and competing explanation***

The proposed transition from altered sensory selection to altered self-experience is an inference. Medford et al. associated depersonalization with reduced insula responses to emotional stimuli and a prefrontal pattern consistent with altered regulation; the small treatment follow-up was neither blinded nor placebo controlled. A single-patient interoception study and a later connectivity study implicate bodily-signal and self-referential processing, without measuring  $\alpha 7$ . (21, 22, 23)

A local loss of inhibition cannot simply be equated to these fMRI findings. One possible bridge is disrupted weighting of bodily versus external signals, followed by altered network regulation; that additional bridge remains a hypothesis, not a demonstrated compensatory response. An equally viable alternative is that anxiety, arousal, sleep disruption, catecholaminergic effects or frontolimbic regulation explain the symptoms without a material  $\alpha 7$  contribution. In a clinical bupropion–paroxetine comparison, the composite containing a DP/DR item showed no differential treatment effect; the composite does not resolve the individual symptom. (24)

### ***Predictions and rejection criteria***

Predictions and rejection criteria for Hypothesis 1 are summarized in Table 2.

Failure of one sensory assay rejects that particular predicted intermediate, not every possible  $\alpha 7$  effect. Conversely, positive P50 or PPI findings alone would not confirm H1. The clinical arm should be observational during independently indicated care; the hypothesis does not require deliberate symptom induction.

#### **4. Hypothesis 2: testing a shared graded mechanism**

##### ***Rationale***

Several studies support protective cholinergic inhibition in particular seizure circuits. Sun et al. found that reducing  $\alpha 7$  activity increased CA1 excitability and shortened pilocarpine seizure latency; their agonist did not significantly improve those outcomes, and the PV interpretation was not established by PV-specific receptor deletion. Wang et al. (2020) used cell-type manipulations to show that a septum–hippocampus cholinergic pathway suppressed mouse seizures through somatostatin-positive interneurons; silencing PV-positive cells did not reverse this protection. A 2025 study linked huperzine-associated seizure protection to septal–dCA1 cholinergic signaling and  $\alpha 7$  pharmacology. These are complementary experiments, not one continuous demonstration of a bupropion–PV– $\alpha 7$  mechanism. (25, 26, 27)

Drug-specific seizure evidence also exists. Silverstone et al. found greater convulsive effects of tested hydroxy, threo and erythro preparations than parent in an acute mouse comparison. The doses were equimolar bupropion-HCl equivalents, individual enantiomers were not separated, and the parent  $CD_{50}$  estimate extended beyond the tested range. The experiment neither identifies  $\alpha 7$  nor provides a human metabolite-risk ranking. Tutka et al. found lower-dose protection against electroshock and higher-dose convulsions, showing that even the parent drug's seizure effects depend on the model. (28, 29)

Human overlap between dissociation and seizures provides motivation, not the molecular explanation. Heydrich et al. studied nine patients with ictal depersonalization-like experiences and seven with derealization-like experiences. The former were predominantly associated with frontal seizure networks and the latter with temporal networks. This selected focal-epilepsy

cohort does not show that primary DPDR is epileptic or that bupropion symptoms progress along a seizure continuum. (30)

***Why a universal continuum is not supported***

$\alpha 7$  is an excitatory cation channel expressed in multiple cell types. Its net network effect depends on which cells it excites, their connections and the prevailing state. Gain-of-function  $\alpha 7$  mutant mice were more susceptible to nicotine-induced seizures, with MLA suppressing those seizures. In another nicotine model MLA was protective, whereas  $\alpha 7$ -null mice retained similar nicotine seizure dose-response curves. A separate *Chrna7*-deficiency study detected no consistent behavioral or EEG phenotype. (31, 32, 33, 34)

Other results constrain a simple GABA-depletion account. Cytisine and tropisetron reduced recurrent seizures in rat TLE paradigms with pharmacological evidence implicating  $\alpha 7$ , but measured bulk GABA did not increase; this does not exclude changed synaptic inhibition, yet it cannot prove that intermediate. Nicotinic blockade also failed to change some hippocampal seizure-like discharges when restricted to  $\alpha 7$ , while broader mecamylamine blockade was effective. (35, 36, 37)

The shared hypothesis consequently requires three demonstrated conditions: relevant  $\alpha 7$  inhibition by the drug, a net loss of effective inhibition in the studied circuit, and an  $\alpha 7$ -dependent relationship between the proposed sensory disturbance and seizure susceptibility. The **strong graded version** predicts that both outcomes covary with the same intermediate within a matched preparation or longitudinal setting. If these conditions fail, separate mechanisms are the better explanation. Dissociation and convulsive seizure measures in different studies cannot by themselves establish an ordered trajectory.

### ***Discriminating predictions***

Discriminating predictions for Hypothesis 2 are summarized in Table 3.

An  $\alpha 7$  agonist “rescue” alone is insufficient: it could improve network behavior through a parallel pathway, have other targets, or change receptor desensitization. Use convergent pharmacological and cell-specific evidence, controls for the rescue agent alone, and comparison of the drug-induced change relative to each preparation's baseline. Lifelong knockout can alter development, so failure or apparent rescue in that model requires separate interpretation.

Animals can test sensory physiology and seizure susceptibility; human subjective depersonalization cannot be inferred from their behavior alone.

### **5. Exposure and alternative targets constrain both hypotheses**

Published rat studies distinguish free extracellular exposure from tissue accumulation. Cremers et al. measured microdialysis concentrations and fitted brain/plasma unbound ratios of 1.9 for parent and 1.7 for hydroxybupropion; their human values were predictions. Bhattacharya et al. found time-varying, stereoselective ratios using a different approach. A 2024 rat study estimated a parent whole-brain unbound/plasma ratio of  $1.94 \pm 0.57$  while separately assessing intracellular accumulation. These observations justify studying brain disposition; they do not demonstrate sufficient human  $\alpha 7$  inhibition. No new exposure model, dose conversion or occupancy estimate is produced here. (38, 39, 40)

Competing pharmacology is substantial. Parent and hydroxybupropion affect 5-HT<sub>3</sub> receptors; bupropion also affected  $\alpha 4\beta 2$  and potassium channels in a human-channel/network screen. TRESK inhibition was measured separately. Conversely, bupropion was an exception to GIRK inhibition reported for several other antidepressants. Thus even “potassium-channel

blockade” requires a named channel and assay. No individual channel hit demonstrates the cause of human seizures. (41, 42, 43, 44, 45)

Both hypotheses are more constrained for routine treatment-associated symptoms than for high-exposure toxicity: the relevant free-concentration premise must be established separately for each setting. Toxicology, therapeutic use and disease-associated circuit alterations are not interchangeable exposure conditions.

## **6. What the completed docking study contributes**

The preserved study used human  $\alpha 7$  templates 8V82 (activated) and 8V8A (desensitized), four regions per state, three seeds, and Vina 1.2.7 with exhaustiveness 32. Its 22 species comprised 20 metabolites or structurally qualified candidates and two parent references, represented by 28 forms. There were 672 metabolite/parent calculations plus 12 shared controls. Chemical forms and uncertain glucuronide linkage were explicitly qualified; no physiological protonation fractions were assigned. The full methods and provenance remain in the saved combined report.

In the matched mid-pore searches, R,R-hydroxybupropion produced pore-lumen rank-1 placements in 12/12 runs across two forms and two states; S,S-hydroxybupropion produced 8/12. Parent rank-1 placements in those boxes landed at the lateral I34 pocket. Earlier parent-only searches also produced pore poses, so the contrast is sensitive to setup. These counts are descriptive computational outcomes, not biological replicates, site preferences, affinity rankings or evidence that R,R is the stronger  $\alpha 7$  antagonist.

Three of four ligand/receptor control pairs recovered the deposited geometry at rank 1. I34/8V82 failed at rank 1 (6.372–6.411 Å) but recovered at rank 2 (0.744–0.783 Å). There was no experimental pore-bound blocker control. Thirteen rank-1 poses and 93 saved modes carried the specified short-contact flag. All alternatives and flags remain available; the loss of 12 early

interrupted-attempt logs/receipts is disclosed in the included incident record. No settings were retuned to turn the failed control into a success.

The usable structural prediction is therefore: **test whether either hydroxybupropion stereoisomer functionally inhibits human  $\alpha 7$  and whether the candidate pore contacts contribute.** Mutational tests would need expression and gating controls because a pore mutation can change drug sensitivity indirectly. Docking adds specificity to the proposed molecular test; it does not verify any circuit or clinical arrow in H1 or H2.

## **7. Evidence assessment and contribution**

The evidence supporting the principal claims is summarized in Table 4.

These assessments describe directness and limitations; they are not formal GRADE ratings. The contribution is a pair of explicit causal proposals that can be tested separately and together, with stereoisomers, circuit sign and exposure as discriminating variables. It is not a claim to have discovered nicotinic antagonism, established hydroxybupropion  $\alpha 7$  potency, or proven a universal dissociation–epilepsy continuum.

## **8. Review methods, source quality and remaining limits**

The evidence table contains **104 endpoint-specific observations from 84 papers**. Consensus supported discovery; Scite supported citation tracing and access to indexed text; PubMed, public primary articles and prescribing information supplied verification. The search was expanded using explicit title/abstract terms to reduce irrelevant automatic term mapping. Six new PubMed queries returned 208 drug–nicotinic records, one metabolite– $\alpha 7$  record, 38 drug–dissociation/gating records, 108 drug–seizure-mechanism records, 191  $\alpha 7$ –seizure records and zero  $\alpha 7$ –depersonalization/derealization records. The single metabolite– $\alpha 7$  hit concerned a behavioral varenicline task, illustrating why hits need screening. The deduplicated deep metadata

file contains 528 journal records; it is not 528 full-text reads. The 208 drug–nicotinic titles were screened for direct assay leads, with selected relevant primary records followed in detail.

Scite forward graphs for Damaj 2004, Papke 2011 and Vázquez-Gómez 2014 contained 210, 61 and 22 resolved edges; a Vázquez backward graph added 45 reference edges. These overlapping graph counts are neither independent replications nor complete worldwide citation coverage. Supporting-labeled citations often concerned transporters, metabolism or other nicotinic receptors. Primary text was preferred over OCR snippets; one  $\mu\text{M}$ -to-mM extraction error and a conference-page result containing adjacent abstracts were identified. Exact ambiguous conference potencies were not used.

The user-supplied citations were also checked against their actual endpoints. The 2014 brain-injury/amygdala paper reported reduced GABA inhibition alongside **increased**  $\alpha 7$  currents, not  $\alpha 7$  loss. The young-rat visual-cortex paper emphasized  $\alpha 4\beta 2$ -related inhibition and other cholinergic effects. The genetic P50 citation did not demonstrate the positive association initially attributed to it. (46, 47, 17)

A retracted 2019  $\alpha 7$ /seizure paper (PMID 30946925) was excluded; the distinct 2021 Sun paper was evaluated on its own record and was not counted as replication of the retracted article. A Fragile-X result was identified as a preprint and not used as a central evidentiary foundation. Patent results on deuterated bupropion were kept separate from peer-reviewed evidence and human metabolites. Negative searches do not establish novelty, and similar broad nicotinic explanations have appeared in public discussion.

This is a focused critical narrative review, not a registered systematic review or meta-analysis. Source access ranged from original methods/results to primary abstracts. The inaccessible complete Damaj and Bondarev texts, incomplete worldwide indexing, small or

selected clinical samples, cross-species differences and assay heterogeneity limit completeness and causal inference. These limits do not prevent formulation of the hypotheses; they prevent presenting them as demonstrated mechanisms. No new docking, MD, free-energy work, exposure modeling, animal experiment or human challenge was performed for this document.

## References

1. National Library of Medicine (US). DailyMed: Wellbutrin XL—bupropion hydrochloride tablet, extended release [Internet]. Bethesda (MD): National Library of Medicine (US); [cited 2026 Sep 26]. Available from: DailyMed prescribing information.
2. Slemmer JE, Martin BR, Damaj MI. Bupropion is a nicotinic antagonist. *J Pharmacol Exp Ther.* 2000;295(1):321-7. PMID: 10991997.
3. Alkondon M, Albuquerque EX. Nicotinic receptor subtypes in rat hippocampal slices are differentially sensitive to desensitization and early in vivo functional up-regulation by nicotine and to block by bupropion. *J Pharmacol Exp Ther.* 2005;313(2):740-50. doi:10.1124/jpet.104.081232.
4. Papke RL, Trocmé-Thibierge C, Guendisch D, Al Rubaiy SA, Bloom SA. Electrophysiological perspectives on the therapeutic use of nicotinic acetylcholine receptor partial agonists. *J Pharmacol Exp Ther.* 2011;337(2):367-79. doi:10.1124/jpet.110.177485.
5. Vázquez-Gómez E, Arias HR, Feuerbach D, Miranda-Morales M, Mihailescu S, Targowska-Duda KM, et al. Bupropion-induced inhibition of  $\alpha 7$  nicotinic acetylcholine receptors expressed in heterologous cells and neurons from dorsal raphe nucleus and hippocampus. *Eur J Pharmacol.* 2014;740:103-11. doi:10.1016/j.ejphar.2014.06.059.
6. Duarte Y, Rojas M, Canan J, Pérez EG, González-Nilo F, García-Colunga J. Different Classes of Antidepressants Inhibit the Rat  $\alpha 7$  Nicotinic Acetylcholine Receptor by Interacting within the Ion Channel: A Functional and Structural Study. *Molecules.* 2021;26(4):998. doi:10.3390/molecules26040998.

7. Lukas RJ, Muresan AZ, Damaj MI, Blough BE, Huang X, Navarro HA, et al. Synthesis and characterization of in vitro and in vivo profiles of hydroxybupropion analogues: aids to smoking cessation. *J Med Chem.* 2010;53(12):4731-48. doi:10.1021/jm1003232.
8. Damaj MI, Carroll FI, Eaton JB, Navarro HA, Blough BE, Mirza S, et al. Enantioselective effects of hydroxy metabolites of bupropion on behavior and on function of monoamine transporters and nicotinic receptors. *Mol Pharmacol.* 2004;66(3):675-82. doi:10.1124/mol.104.001313.
9. Bondarev ML, Bondareva TS, Young R, Glennon RA. Behavioral and biochemical investigations of bupropion metabolites. *Eur J Pharmacol.* 2003;474(1):85-93. doi:10.1016/s0014-2999(03)02010-7.
10. Albuquerque EX, Pereira EF, Mike A, Eisenberg HM, Maelicke A, Alkondon M. Neuronal nicotinic receptors in synaptic functions in humans and rats: physiological and clinical relevance. *Behav Brain Res.* 2000;113(1-2):131-41. doi:10.1016/s0166-4328(00)00208-4.
11. Ji D, Dani JA. Inhibition and disinhibition of pyramidal neurons by activation of nicotinic receptors on hippocampal interneurons. *J Neurophysiol.* 2000;83(5):2682-90. doi:10.1152/jn.2000.83.5.2682.
12. Pidoplichko VI, Prager EM, Aroniadou-Anderjaska V, Braga MF.  $\alpha 7$ -Containing nicotinic acetylcholine receptors on interneurons of the basolateral amygdala and their role in the regulation of the network excitability. *J Neurophysiol.* 2013;110(10):2358-69. doi:10.1152/jn.01030.2012.
13. Townsend M, Whyment A, Walczak JS, Jeggo R, van den Top M, Flood DG, et al.  $\alpha 7$ -nAChR agonist enhances neural plasticity in the hippocampus via a GABAergic circuit. *J Neurophysiol.* 2016;116(6):2663-2675. doi:10.1152/jn.00243.2016.

14. Olincy A, Harris JG, Johnson LL, Pender V, Kongs S, Allensworth D, et al. Proof-of-concept trial of an  $\alpha 7$  nicotinic agonist in schizophrenia. *Arch Gen Psychiatry*. 2006;63(6):630-8. doi:10.1001/archpsyc.63.6.630.
15. Shiina A, Shirayama Y, Niitsu T, Hashimoto T, Yoshida T, Hasegawa T, et al. A randomised, double-blind, placebo-controlled trial of tropisetron in patients with schizophrenia. *Ann Gen Psychiatry*. 2010;9:27. doi:10.1186/1744-859X-9-27.
16. Winterer G, Gallinat J, Brinkmeyer J, Musso F, Kornhuber J, Thuerauf N, et al. Allosteric  $\alpha$ -7 nicotinic receptor modulation and P50 sensory gating in schizophrenia: a proof-of-mechanism study. *Neuropharmacology*. 2013;64:197-204. doi:10.1016/j.neuropharm.2012.06.040.
17. Bertelsen B, Oranje B, Melchior L, Fagerlund B, Werge TM, Mikkelsen JD, et al. Association Study of CHRNA7 Promoter Variants with Sensory and Sensorimotor Gating in Schizophrenia Patients and Healthy Controls: A Danish Case-Control Study. *Neuromolecular Med*. 2015;17(4):423-30. doi:10.1007/s12017-015-8371-9.
18. Schabinger N, Gillmeister H, Berti S, Michal M, Beutel ME, Adler J. Detached and distracted: ERP correlates of altered attentional function in depersonalisation. *Biol Psychol*. 2018;134:64-71. doi:10.1016/j.biopsycho.2018.02.014.
19. Dale KY, Flaten MA, Elden A, Holte A. Dissociative identity disorder and prepulse inhibition of the acoustic startle reflex. *Neuropsychiatr Dis Treat*. 2008;4(3):653-62. doi:10.2147/ndt.s3006.
20. Mori T, Shibasaki M, Ogawa Y, Hokazono M, Wang TC, Rahmadi M, et al. Comparison of the behavioral effects of bupropion and psychostimulants. *Eur J Pharmacol*. 2013;718(1-3):370-5. doi:10.1016/j.ejphar.2013.07.046.

21. Medford N, Sierra M, Stringaris A, Giampietro V, Brammer MJ, David AS. Emotional Experience and Awareness of Self: Functional MRI Studies of Depersonalization Disorder. *Front Psychol.* 2016;7:432. doi:10.3389/fpsyg.2016.00432.
22. Sedeño L, Couto B, Melloni M, Canales-Johnson A, Yoris A, Baez S, et al. How do you feel when you can't feel your body? Interoception, functional connectivity and emotional processing in depersonalization-derealization disorder. *PLoS One.* 2014;9(6):e98769. doi:10.1371/journal.pone.0098769.
23. Jia Y, Song N, Ning Y, Zhu H, Dong L, Feng S, et al. Altered Self-Referential-Related Brain Regions in Depersonalization-Derealization Disorder. *Brain Behav.* 2025;15(2):e70314. doi:10.1002/brb3.70314.
24. Grunebaum MF, Keilp JG, Ellis SP, Sudol K, Bauer N, Burke AK, et al. SSRI versus bupropion effects on symptom clusters in suicidal depression: post hoc analysis of a randomized clinical trial. *J Clin Psychiatry.* 2013;74(9):872-9. doi:10.4088/JCP.12m08000.
25. Sun P, Liu DG, Ye XM. Nicotinic Acetylcholine Receptor  $\alpha 7$  Subunit Is an Essential Regulator of Seizure Susceptibility. *Front Neurol.* 2021;12:656752. doi:10.3389/fneur.2021.656752.
26. Wang Y, Wang Y, Xu C, Wang S, Tan N, Chen C, et al. Direct Septum-Hippocampus Cholinergic Circuit Attenuates Seizure Through Driving Somatostatin Inhibition. *Biol Psychiatry.* 2020;87(9):843-856. doi:10.1016/j.biopsych.2019.11.014.
27. Wang Y, Hu KY, Zhang QY, Song YJ, Li LJ, Wang F, et al. Huperzine A attenuates epileptic seizures via enhancing dCA1-projecting septal cholinergic transmission. *Acta Pharmacol Sin.* 2025;46(8):2151-2162. doi:10.1038/s41401-025-01522-w.

28. Silverstone PH, Williams R, McMahon L, Fleming R, Fogarty S. Convulsive liability of bupropion hydrochloride metabolites in Swiss albino mice. *Ann Gen Psychiatry*. 2008;7:19. doi:10.1186/1744-859X-7-19.
29. Tutka P, Barczynski B, Wielosz M. Convulsant and anticonvulsant effects of bupropion in mice. *Eur J Pharmacol*. 2004;499(1-2):117-20. doi:10.1016/j.ejphar.2004.07.105.
30. Heydrich L, Marillier G, Evans N, Seeck M, Blanke O. Depersonalization- and derealization-like phenomena of epileptic origin. *Ann Clin Transl Neurol*. 2019;6(9):1739-1747. doi:10.1002/acn3.50870.
31. Gil Z, Sack RA, Kedmi M, Harmelin A, Orr-Urtreger A. Increased sensitivity to nicotine-induced seizures in mice heterozygous for the L250T mutation in the  $\alpha 7$  nicotinic acetylcholine receptor. *Neuroreport*. 2002;13(2):191-6. doi:10.1097/00001756-200202110-00004.
32. Iha HA, Kunisawa N, Shimizu S, Tokudome K, Mukai T, Kinboshi M, et al. Nicotine Elicits Convulsive Seizures by Activating Amygdalar Neurons. *Front Pharmacol*. 2017;8:57. doi:10.3389/fphar.2017.00057.
33. Franceschini D, Paylor R, Broide R, Salas R, Bassetto L, Gotti C, et al. Absence of  $\alpha 7$ -containing neuronal nicotinic acetylcholine receptors does not prevent nicotine-induced seizures. *Brain Res Mol Brain Res*. 2002;98(1-2):29-40. doi:10.1016/s0169-328x(01)00309-6.
34. Yin J, Chen W, Yang H, Xue M, Schaaf CP. *Chrna7* deficient mice manifest no consistent neuropsychiatric and behavioral phenotypes. *Sci Rep*. 2017;7:39941. doi:10.1038/srep39941.

35. Zheng JJ, Zhang TY, Liu HT, Huang ZX, Teng JM, Deng JX, et al. Cytisine Exerts an Anti-Epileptic Effect via  $\alpha 7$ nAChRs in a Rat Model of Temporal Lobe Epilepsy. *Front Pharmacol.* 2021;12:706225. doi:10.3389/fphar.2021.706225.
36. Qian X, Sheng X, Ding J, Yiming Z, Zheng J, Zhong J, et al. Tropisetron, an Antiemetic Drug, Exerts an Anti-Epileptic Effect Through the Activation of  $\alpha 7$ nAChRs in a Rat Model of Temporal Lobe Epilepsy. *CNS Neurosci Ther.* 2024;30(10):e70086. doi:10.1111/cns.70086.
37. Zapukhliak O, Netsyk O, Romanov A, Maximyuk O, Oz M, Holmes GL, et al. Mecamylamine inhibits seizure-like activity in CA1-CA3 hippocampus through antagonism to nicotinic receptors. *PLoS One.* 2021;16(3):e0240074. doi:10.1371/journal.pone.0240074.
38. Cremers TI, Flik G, Folgering JH, Rollema H, Stratford RE. Development of a Rat Plasma and Brain Extracellular Fluid Pharmacokinetic Model for Bupropion and Hydroxybupropion Based on Microdialysis Sampling, and Application to Predict Human Brain Concentrations. *Drug Metab Dispos.* 2016;44(5):624-33. doi:10.1124/dmd.115.068932.
39. Bhattacharya CS, Masters AR, Bach C, Sinn AL, Trowbridge MA, Stratford RE. Characterization of the Stereoselective Disposition of Bupropion and Its Metabolites in Rat Plasma and Brain. *Eur J Drug Metab Pharmacokinet.* 2023;48(2):171-187. doi:10.1007/s13318-023-00817-9.
40. Bällgren F, Hu Y, Li S, van de Beek L, Hammarlund-Udenaes M, Loryan I. Region-independent active CNS net uptake of marketed H<sup>+</sup>/OC antiporter system substrates. *Front Cell Neurosci.* 2024;18:1493644. doi:10.3389/fncel.2024.1493644.

41. Pandhare A, Pappu AS, Wilms H, Blanton MP, Jansen M. The antidepressant bupropion is a negative allosteric modulator of serotonin type 3A receptors. *Neuropharmacology*. 2017;113(Pt A):89-99. doi:10.1016/j.neuropharm.2016.09.021.
42. Stuebler AG, Jansen M. Bupropion Inhibits Serotonin Type 3AB Heteromeric Channels at Clinically Relevant Concentrations. *Mol Pharmacol*. 2020;97(3):171-179. doi:10.1124/mol.119.118349.
43. Rockley K, Roberts R, Jennings H, Jones K, Davis M, Levesque P, et al. An integrated approach for early in vitro seizure prediction utilizing hiPSC neurons and human ion channel assays. *Toxicol Sci*. 2023;196(1):126-140. doi:10.1093/toxsci/kfad087.
44. Park H, Kim EJ, Han J, Han J, Kang D. Effects of analgesics and antidepressants on TREK-2 and TRESK currents. *Korean J Physiol Pharmacol*. 2016;20(4):379-85. doi:10.4196/kjpp.2016.20.4.379.
45. Kobayashi T, Washiyama K, Ikeda K. Inhibition of G protein-activated inwardly rectifying K<sup>+</sup> channels by various antidepressant drugs. *Neuropsychopharmacology*. 2004;29(10):1841-51. doi:10.1038/sj.npp.1300484.
46. Almeida-Suhett CP, Prager EM, Pidoplichko V, Figueiredo TH, Marini AM, Li Z, et al. Reduced GABAergic inhibition in the basolateral amygdala and the development of anxiety-like behaviors after mild traumatic brain injury. *PLoS One*. 2014;9(7):e102627. doi:10.1371/journal.pone.0102627.
47. Lucas-Meunier E, Monier C, Amar M, Baux G, Frégnac Y, Fossier P. Involvement of nicotinic and muscarinic receptors in the endogenous cholinergic modulation of the balance between excitation and inhibition in the young rat visual cortex. *Cereb Cortex*. 2009;19(10):2411-27. doi:10.1093/cercor/bhn258.

**Table 1.** Comparative inhibition of non- $\alpha 7$  nicotinic receptor subtypes by bupropion and hydroxybupropion stereoisomers

| Compound             | $\alpha 4\beta 2$ IC <sub>50</sub> , $\mu$ M | $\alpha 3\beta 4^*$ IC <sub>50</sub> , $\mu$ M | $\alpha 7$ inference                 |
|----------------------|----------------------------------------------|------------------------------------------------|--------------------------------------|
| Bupropion            | 12                                           | 1.8                                            | Separate parent experiments required |
| S,S-hydroxybupropion | 3.3                                          | 11                                             | Not assayed in this panel            |
| R,R-hydroxybupropion | 31                                           | 6.5                                            | Not assayed in this panel            |

Values are central estimates from the same assay; the source reports multiplicative SEM. The asterisk allows additional subunits. Neither “hydroxybupropion is more potent at  $\alpha 7$ ” nor “hydroxybupropion is weaker at  $\alpha 7$ ” follows from this table. Damaj's primary abstract supports stereoselective  $\alpha 4\beta 2$  pharmacology; Bondarev's abstract supports tested-metabolite  $\alpha 3\beta 4$  functional inhibition. Complete original text for those two papers remained inaccessible in this review, and exact unverified entries were not filled in from secondary citations. (8, 9)

**Table 2.** Testable predictions and rejection criteria for Hypothesis 1

| Testable prediction                                                   | Evidence that would support H1                                                                                                               | Evidence that would weaken or reject the proposed route                                                                              |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Direct target action precedes the circuit effect                      | Parent or an identified stereoisomer inhibits human $\alpha 7$ within a justified unbound exposure range                                     | No relevant inhibition; symptoms and circuit effects remain at exposures below the demonstrated range                                |
| $\alpha 7$ contributes specifically to altered inhibitory recruitment | Drug changes in identified interneuron output diminish when $\alpha 7$ engagement is selectively prevented, with baseline function preserved | Effects persist despite verified loss of the drug's $\alpha 7$ interaction and are explained by matched non- $\alpha 7$ pharmacology |
| The proposed sensory intermediate relates to symptoms                 | Within-person changes in a prespecified sensory measure track independently measured dissociation after accounting for confounds             | Sensory changes occur without symptoms, or symptoms track another mechanism without the predicted intermediate                       |
| DP and DR are measured separately                                     | Replicable symptom-specific associations with the predicted circuit changes                                                                  | Only a broad distress or composite score changes                                                                                     |

H1, Hypothesis 1; DP, depersonalization; DR, derealization.

**Table 3.** Discriminating predictions for the shared-mechanism hypothesis

| Proposed observation                                                                                                                        | Interpretation                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| A drug-induced loss of $\alpha 7$ -dependent inhibitory recruitment predicts both sensory disruption and reduced seizure threshold          | Supports the shared pathway, subject to exposure and specificity checks                                   |
| A selective intervention preventing the drug– $\alpha 7$ interaction reduces both outcomes without merely suppressing baseline excitability | Stronger evidence for mediation than a general anticonvulsant rescue                                      |
| Sensory effects are $\alpha 7$ -dependent, while seizure effects persist after $\alpha 7$ -specific protection                              | H1 remains viable; the shared version of H2 is weakened or rejected                                       |
| Seizure susceptibility changes without the proposed sensory intermediate                                                                    | Supports a seizure-specific pathway or competing mechanism, not a required sensory-to-seizure progression |
| Different stereoisomers separate the outcomes after controlling for exposure and other targets                                              | Demonstrates mechanistic heterogeneity; tests rather than assumes the metabolite contribution             |

H1, Hypothesis 1; H2, Hypothesis 2.

**Table 4.** Assessment of the principal claims after literature review

| Claim                                                                                     | Assessment after review                                                             |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Parent bupropion inhibits $\alpha 7$ in experimental preparations                         | Directly supported; potency depends on assay                                        |
| Each human metabolite inhibits $\alpha 7$ , with hydroxybupropion more potent than parent | Not established in the verified primary records                                     |
| $\alpha 7$ can recruit inhibition and influence seizure susceptibility                    | Supported in specific circuits; opposite and null results also exist                |
| Bupropion causes dissociation through $\alpha 7$                                          | Hypothesis; target exposure and symptom mediation unverified                        |
| Dissociative symptoms and seizures share a graded $\alpha 7$ mechanism                    | Stronger hypothesis requiring joint evidence; not established by separate endpoints |
| Docking demonstrates binding or affinity                                                  | Unsupported; the outputs are candidate geometries                                   |